CASE B

Ms. Elena Rodriguez | 24F, Postgraduate student

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Acute bronchoconstriction — ↑ airway resistance, ↓ expiratory flow
- Unable to speak in full sentences
- Dysfunctional breathing pattern — upper chest, accessory muscle dominant
- Cannot perform ADLs; completely activity-limited
- Possible mucus hypersecretion and plugging
- Unable to attend academic commitments
- Suboptimal inhaler technique — reliever not working
- Anxiety and breathlessness-fear cycle perpetuating symptoms
- Likely poor ICS preventer adherence — poorly controlled baseline
- Risk of recurrent ED presentations without improved self-management

■ RED FLAGS

- Red Flags (5)
- Silent Chest No wheeze despite severe SOB — air movement so reduced it cannot generate wheeze. Pre-respiratory arrest → EMERGENCY — call resuscitation team immediately; do not proceed
- Silent Chest
- No wheeze despite severe SOB — air movement so reduced it cannot generate wheeze. Pre-respiratory arrest
- → EMERGENCY — call resuscitation team immediately; do not proceed
- SpO■ <92% / cyanosis Severe hypoxaemia from V/Q mismatch; impending respiratory failure → Escalate O■; urgent medical review; prepare for possible intubation
- SpO■ <92% / cyanosis
- Severe hypoxaemia from V/Q mismatch; impending respiratory failure

■ PATHOPHYSIOLOGY

- Asthma is characterised by chronic eosinophilic airway inflammation and bronchial hyperresponsiveness (BHR).
- In sensitised individuals (atopic asthma), airway epithelium is primed to over-react to triggers.
- Rodriguez has two simultaneous triggers: viral URTI causes direct epithelial injury, releasing inflammatory mediators (histamine, leukotrienes, prostaglandins) that amplify BHR; pollen allergens cross-link IgE on mast cells → degranulation → acute bronchoconstriction.
- Bronchoconstriction dramatically increases airway resistance.
- By Poiseuille's law, resistance ∝ 1/r■ — halving the airway radius increases resistance 16-fold.

■ DIFFERENTIAL DIAGNOSES

1. Vocal Cord Dysfunction (VCD) / Inducible Laryngeal Obstruction (ILO)

- Young female — VCD is most prevalent in young women aged 20–40
- Episodic breathlessness — can be triggered by stress or exercise (postgraduate student, high pressure environment)
- Does not fully respond to bronchodilators

2. Anaphylaxis with Bronchospasm

- Atopic individual with acute onset severe breathlessness
- High-pollen allergen exposure on day of presentation
- Bronchospasm can be a component of a systemic allergic reaction

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Breathing Pattern Assessment

Test 1: Breathing Pattern Assessment What it measures Identifies dysfunctional breathing patterns — upper chest dominance, hyperventilation, irregular rhythm, mouth breathing — that worsen bronchoconstriction and maintain the acute episode. Directly guides the breathing retrainin

Inhaler Technique Assessment (Observed)

Test 2: Inhaler Technique Assessment (Observed) What it measures Identifies specific, correctable errors in pMDI or DPI technique — a modifiable cause of poor asthma control. Up to 70–80% of patients demonstrate incorrect technique. How to perform Ask patient to demonstrate using

Chest Auscultation (Post-Bronchodilator)

Test 3: Chest Auscultation (Post-Bronchodilator) What it measures Assesses response to bronchodilator therapy; monitors for silent chest (emergency indicator); identifies any secretion presence requiring airway clearance. How to perform Patient seated upright if tolerated; expose

■ TREATMENT PLAN

■ Initial Phase — Acute Management (ED / Hours 1–48)

- Shake the inhaler; remove the cap
- Breathe out gently (not forcefully)
- Seal lips firmly around mouthpiece (or spacer mouthpiece)
- Begin slow inhalation — press canister down once at the very start
- Continue slow inhalation over 3–5 seconds

■ Patient Education — Key Messages

• Preventer vs. reliever — the most important distinction: "Your preventer inhaler [ICS] must be taken every day — even when you feel completely well. It reduces the inflammation that causes these attacks. Your blue inhaler is for relief in the moment — it does not treat the underlying inflammation."

• Written Asthma Action Plan: Traffic-light system — Green (well, take preventer as usual); Amber (increasing symptoms — take extra reliever, start oral steroid if prescribed, contact GP); Red (severe attack not responding — call 999 immediately). Every patient should leave hospital with a written plan.

• Trigger avoidance: Check daily pollen count (high pollen days — keep windows closed, shower after being outdoors, wear wrap-around sunglasses). For viral illness — begin reliever use early at first sign of symptoms; contact GP proactively.

• When to call emergency services: "If your blue inhaler is not working after 4 puffs in 10 minutes, your breathing is getting worse, or you cannot speak — call 999. Never wait to see if it improves on its own in that situation."

■ Long-Term — Weeks 1–4 Post-Discharge and Self-Management

• GP/Asthma nurse follow-up within 48 hours of discharge: Review medication step; ensure written action plan provided; consider specialist referral if this is a second ED attendance within 12 months.

• Breathing retraining programme: Buteyko or Papworth method — evidence supports ↓ reliever use and ↑ quality of life in chronic dysfunctional breathing. Particularly appropriate given the likely anxiety-asthma overlap.

• Psychological support: Screen for health anxiety and panic disorder (both share positive feedback loops with breathlessness). Refer for CBT if indicated — stress management also reduces exacerbation frequency.

• Discharge criteria: Demonstrates correct inhaler technique independently; has personalised written action plan; peak flow ≥80% personal best; ACT score ≥20 (well-controlled); follow-up appointment within 1 week confirmed.